

MANGO, Togo

WMO: 653520

Lat: 10.367N

Lon: 0.467E

Elev: 479

StdP: 14.44

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	65.8	67.7	25.4	19.8	81.9	27.9	22.2	81.9	11.9	80.7	10.8	81.2	4.7	N/A	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	20.5	103.8	74.2	102.3	74.3	100.7	74.4	82.6	92.6	81.7	92.1	81.0	91.5	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	159.9	87.4	79.2	154.3	86.6	78.5	150.7	86.0	46.8	92.4	45.9	92.7	45.1	91.8	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	9.7	8.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			48.1	85.3	1.6	1.6	47.0	86.4	46.0	87.3	45.2	88.2	44.0	89.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.0	81.2	87.0	91.0	89.6	86.1	83.2	80.7	79.9	80.8	83.1	83.8	81.3
	DBStd	4.51	3.37	3.35	2.81	3.39	3.41	3.09	2.49	2.24	2.47	1.97	2.19	2.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12396	967	1035	1270	1189	1119	996	953	928	925	1027	1015	972
	CDD65	6921	502	615	805	739	654	546	488	463	475	562	565	507
	CDH74	88783	6386	9257	12886	11432	9123	6661	4901	4253	4597	6112	7032	6143
CDH80	48075	3500	5725	8528	7253	5018	3085	1758	1341	1682	2840	3996	3349	
Wind	WSAvg	3.0	4.6	3.6	2.9	3.7	3.1	2.9	2.4	2.2	1.8	1.7	2.4	4.3
Precipitation	PrecAvg	42.9	0.0	0.1	0.8	2.4	4.1	5.7	8.1	9.3	8.1	2.9	0.1	0.1
	PrecMax	62.2	1.1	1.5	4.7	8.2	9.1	10.3	16.5	15.8	13.9	7.5	1.7	1.9
	PrecMin	35.5	0.0	0.0	0.0	0.1	1.0	2.1	3.3	3.7	2.7	0.1	0.0	0.0
	PrecStd	5.6	0.2	0.3	0.9	1.8	1.6	2.0	3.3	2.8	2.8	1.7	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.7	105.2	106.0	104.7	100.8	97.1	92.9	91.2	93.4	97.0	100.4	100.3
		MCWB	68.4	73.2	73.1	76.6	80.1	79.2	79.7	79.1	80.6	78.3	73.9	69.4
	2%	DB	98.3	102.4	104.1	103.1	98.7	95.0	90.9	89.4	91.1	94.9	98.6	97.9
		MCWB	67.5	71.8	73.7	77.0	79.8	79.3	79.5	79.1	80.0	78.8	73.8	69.2
	5%	DB	96.0	100.4	102.6	101.3	97.0	93.1	89.4	87.9	89.5	93.1	97.0	95.9
		MCWB	65.5	70.1	73.7	78.0	79.6	79.5	79.2	79.0	79.9	78.9	73.9	67.7
	10%	DB	93.6	98.8	101.3	99.6	95.3	91.5	87.8	86.3	87.8	91.2	95.3	93.5
		MCWB	64.3	69.3	73.9	78.7	79.6	79.4	78.8	78.7	79.4	79.1	73.7	66.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.4	79.3	81.9	84.7	83.4	83.2	82.2	81.3	82.7	82.8	80.9	78.4
		MCDB	94.6	98.7	98.7	95.9	93.6	91.1	88.5	87.9	89.5	89.9	94.2	94.3
	2%	WB	71.7	77.4	80.0	81.9	81.9	81.6	80.7	80.2	81.5	81.8	79.4	75.8
		MCDB	92.0	96.8	97.1	96.0	93.5	90.9	88.2	86.9	88.6	89.5	93.3	91.6
	5%	WB	68.2	75.8	78.9	81.0	81.1	80.7	79.9	79.5	80.6	80.9	78.2	72.6
		MCDB	92.4	93.6	96.4	95.6	93.1	89.9	87.1	86.0	87.8	89.4	92.1	88.6
	10%	WB	66.0	74.2	78.0	80.3	80.4	79.9	79.1	78.9	79.8	80.1	77.1	69.1
		MCDB	91.4	91.9	95.0	94.9	92.5	88.8	86.1	85.1	86.6	88.7	90.4	90.5
Mean Daily Temperature Range	5% DB	MDBR	25.0	23.6	20.5	18.0	15.9	14.7	12.6	11.9	13.7	18.1	26.6	26.7
		MCDBR	29.0	25.8	22.8	20.6	18.3	17.5	15.6	14.7	16.5	20.8	28.1	29.9
		MCWBR	12.7	11.7	8.9	6.5	5.5	5.6	5.6	5.7	7.0	7.4	9.3	12.5
	5% WB	MCDBR	27.0	21.9	19.4	17.8	16.7	15.7	14.3	13.3	15.6	17.9	24.3	27.3
		MCWBR	14.1	11.6	7.6	6.2	6.4	6.3	6.2	6.0	7.4	7.8	9.1	14.2
Clear-Sky Solar Irradiance	taub	0.679	0.742	0.812	0.721	0.631	0.579	0.528	0.514	0.521	0.605	0.599	0.598	
	taud	1.635	1.535	1.450	1.655	1.790	1.924	2.103	2.145	2.100	1.856	1.805	1.788	
	Ebn at Noon	209	202	192	209	224	233	246	253	254	231	227	225	
	Edh at Noon	81	92	102	82	70	61	51	49	52	66	68	68	
All-Sky Solar Radiation	RadAvg	1718	1784	1847	1895	1840	1685	1519	1453	1583	1797	1767	1692	
	RadStd	73	68	82	66	64	52	54	62	60	59	37	59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page